

Chap 4. Entanglement

H^A, H^B : Hilbert sp.

$$D(H) = \{ \rho \in L(H) : \rho^* = \rho, \rho \geq 0, \text{tr}(\rho) = 1 \}$$

$\hookrightarrow$ density op.

$$H^A \otimes H^B = \{ \psi : H^A \times H^B \longrightarrow \mathbb{C} : \text{anti-lin \& cont.} \}$$

Def $\rho \in D(H^A \otimes H^B)$ is "separable" w.r.t H^A & H^B

$$\Leftrightarrow \exists \rho_j^A \in D(H^A), \rho_j^B \in D(H^B), \rho_j \geq 0$$

s.t. $\sum_j \rho_j = 1$ & $\rho = \sum_j \rho_j \rho_j^A \otimes \rho_j^B$

O.W. ρ is "entangled".

Thm A pure state $|\psi\rangle \in H^A \otimes H^B$ is separable

$$\Leftrightarrow \exists \text{ pure } |\phi\rangle \in H^A \text{ and pure } |\chi\rangle \in H^B \text{ s.t.}$$
$$|\psi\rangle = |\phi\rangle \otimes |\chi\rangle$$

O.W. $|\psi\rangle$ is entangled.

e.g. $|\psi\rangle = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle)$

$$= \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Thm For pure $|\psi\rangle \in H^A \otimes H^B$

$$|\psi\rangle : \text{separable} \Leftrightarrow \rho^A(\psi) \& \rho^B(\psi) : \text{separable.}$$

Equivalently,

$$|\psi\rangle : \text{entangled} \Leftrightarrow 1 \text{ of them : true mixture.}$$

$\rightarrow$ identical H.s.

Def A pure $|\Psi\rangle \in \mathbb{H}^A \otimes \mathbb{H}^A$ is "maximally entangled"
 $\Leftrightarrow \rho^A(\Psi) = \lambda \mathbb{1}$ with $0 < \lambda < 1$.
 $\hookrightarrow$ identity mtr

e.g Bell basis $\{|\Phi^\pm\rangle, |\Psi^\pm\rangle\}$

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$= \frac{1}{\sqrt{2}} (|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$$

$\hookrightarrow \Phi_{00}^+, \Phi_{11}^+$

$$\rho^A(\Phi) = \sum_{a_1, a_2} \Phi_{a_1, a_2}^+ \Phi_{a_1, a_2}^+ |e_{a_1}\rangle \langle e_{a_2}|$$

$$= \sum_{a_1, a_2} (\Phi_{a_1, 0}^+ \overline{\Phi_{a_1, 0}^+} + \Phi_{a_1, 1}^+ \overline{\Phi_{a_1, 1}^+}) |e_{a_1}\rangle \langle e_{a_2}|$$

$$= \frac{1}{2} |0\rangle \langle 0| + \frac{1}{2} |1\rangle \langle 1| = \frac{1}{2} \mathbb{1}^A.$$

③ Entanglement Swapping.

"system can be entangled even if they have not interacted"

$$|\Phi\rangle^{ABCD} \in \mathbb{H}^A \otimes \mathbb{H}^B \otimes \mathbb{H}^C \otimes \mathbb{H}^D =: \mathbb{H}^{ABCD}$$

$$|\Phi\rangle^{ABCD} = |\Psi^-\rangle^{AB} \otimes |\Psi^-\rangle^{CD}$$

$$= \frac{1}{2} (|0101\rangle - |0110\rangle - |1001\rangle + |1010\rangle)$$

$$= \frac{1}{2} (|\Psi^+\rangle^{AD} \otimes |\Psi^+\rangle^{BC} - |\Psi^+\rangle^{AD} \otimes |\Psi^-\rangle^{BC}$$

$$- |\Phi^+\rangle^{AD} \otimes |\Phi^+\rangle^{BC} + |\Phi^-\rangle^{AD} \otimes |\Phi^-\rangle^{BC})$$

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\Sigma_z^{BC} := 1 \otimes \sigma_z \otimes \sigma_z \otimes 1$$

$$\Sigma_x^{BC} := 1 \otimes \sigma_x \otimes \sigma_x \otimes 1.$$

$$\text{Since } [\sigma_z \otimes \sigma_z, \sigma_x \otimes \sigma_x] = 0, [\Sigma_z^{BC}, \Sigma_x^{BC}] = 0.$$

$\Rightarrow$ compatible $\Rightarrow$ yield sharp results. i.e. no uncertainty.

$$\text{e.g. } (\Sigma_z^{BC}, \Sigma_x^{BC}) = (-1, +1)$$

$\Rightarrow$ particle-pair BC: $|\psi^+\rangle^{BC}$ (see Table 4.1

Also " AD: $|\psi^+\rangle^{AD}$. in the book)

① E-P-R Paradox

$\hookrightarrow$ "Q.M is incomplete".

Goal (To show): Two claims are wrong.

EPR claim 1 Quantum mechanics is complete.

complete system

elt of the physical reality of the system

$\Uparrow$ corresp

elt in the physical theory.

elt of reality of system with the value of this quantity can be predicted with certainty.

e.g. $|0\rangle$: eig vec of σ_z with eig.val $+1$.

$\Rightarrow$ value of the physical quantity 'spin in z-dir' is $+1$.

$\Rightarrow$ elt of reality.

But $|0\rangle$ is not an eig. vec. of σ_x .

$$\Rightarrow \Delta_{|0\rangle}(\sigma_x) = 1$$

$\hookrightarrow$ uncertainty.

$M_1 M_2 \neq M_2 M_1 \Rightarrow$ eig. vec. not same.

$\Rightarrow$ cannot predict with certainty.

EPR claim 2 The values of incompatible observables are not jointly real.

$$C1 \Rightarrow C2.$$

⊖ Assume $\sim C2$.

(If $C1$ holds, state gives certain prediction.
* (⊖ $\sim C2$))

By using entangled states, E-P-R ^{also} showed $C1 \Rightarrow \sim C2$.
 $\rightarrow$ "EPR Paradox".

Claim 1 is False.

$$\begin{aligned} |\Xi^+\rangle &= \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}} \left(\underbrace{|\uparrow_z\rangle}_{\text{Alice}} \otimes \underbrace{|\uparrow_z\rangle}_{\text{Bob}} + \underbrace{|\downarrow_z\rangle}_A \otimes \underbrace{|\downarrow_z\rangle}_B \right) \\ &= \frac{1}{\sqrt{2}} (|\uparrow_z\rangle \otimes |\uparrow_z\rangle + |\downarrow_z\rangle \otimes |\downarrow_z\rangle). \end{aligned}$$

$$\begin{aligned} \sigma_z \otimes \mathbb{1} &\rightarrow \text{Eig}(\sigma_z, +1) = \text{span}\{ |\uparrow_z\rangle \otimes |\psi\rangle : |\psi\rangle \in \mathbb{H}^B \} \\ &\quad \text{Eig}(\sigma_z, -1) = \quad \quad \quad " \quad |\downarrow_z\rangle \quad \quad \quad " \end{aligned}$$

$$P_{z,t+1} = |\uparrow_z\rangle\langle\uparrow_z| \otimes \mathbb{1}, \quad P_{z,-1} = |\downarrow_z\rangle\langle\downarrow_z| \otimes \mathbb{1}.$$

$$\begin{aligned} P_{z,t+1} |\Phi^+\rangle &= \frac{1}{\sqrt{2}} (|\uparrow_z\rangle\langle\uparrow_z| \otimes \mathbb{1}) (|\uparrow_z\rangle \otimes |\uparrow_z\rangle + |\downarrow_z\rangle \otimes |\downarrow_z\rangle) \\ &= \frac{1}{\sqrt{2}} (|\uparrow_z\rangle\langle\uparrow_z|\uparrow_z\rangle \otimes |\uparrow_z\rangle + |\uparrow_z\rangle\langle\uparrow_z|\downarrow_z\rangle \otimes |\downarrow_z\rangle) \\ &= \frac{1}{\sqrt{2}} |\uparrow_z\rangle \otimes |\uparrow_z\rangle \end{aligned}$$

$$P_{z,-1} |\Phi^+\rangle = \frac{1}{\sqrt{2}} |\downarrow_z\rangle \otimes |\downarrow_z\rangle$$

$$|\bar{\psi}_{z,t+1}\rangle := \frac{P_{z,t+1} |\Phi^+\rangle}{\|P_{z,t+1} |\Phi^+\rangle\|} = \frac{(|\uparrow_z\rangle\langle\uparrow_z| \otimes \mathbb{1}) |\Phi^+\rangle}{\|(|\uparrow_z\rangle\langle\uparrow_z| \otimes \mathbb{1}) |\Phi^+\rangle\|}.$$

$$= |\uparrow_z\rangle \otimes |\uparrow_z\rangle.$$

$$\begin{aligned} \rho^b(|\bar{\psi}_{z,t+1}\rangle\langle\bar{\psi}_{z,t+1}|) &= \text{tr}^A(|\bar{\psi}_{z,t+1}\rangle\langle\bar{\psi}_{z,t+1}|) \\ &= \text{tr}^A(|\uparrow_z\rangle\langle\uparrow_z| \otimes |\uparrow_z\rangle\langle\uparrow_z|) \\ &= \text{tr}^A(|\uparrow_z\rangle\langle\uparrow_z| \otimes |\uparrow_z\rangle\langle\uparrow_z|) \\ &= \text{tr}(|\uparrow_z\rangle\langle\uparrow_z|) |\uparrow_z\rangle\langle\uparrow_z| = \boxed{|\uparrow_z\rangle\langle\uparrow_z|}. \end{aligned}$$